İlhan Bahadır Yavaş

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Software Engineer and Game Developer with a strong foundation in C#, C++, and Object-Oriented Programming. Experienced in developing interactive gameplay mechanics, optimizing game loops, and building multiplayer network architectures using Unity and Unreal Engine. Adept at rapid prototyping for 2D/3D games and applying Agile/Scrum methodologies to deliver polished player experiences. Passionate about problem-solving and writing clean, scalable code for dynamic game environments.

EDUCATION

İhsan Doğramacı Bilkent University — **Ankara**

İnformation Systems and Technologies — 2021-2026

WORK EXPERIENCE

INFINIA Engineering | Unreal Engine Game Developer | Full-Time | Ankara — Jan 2025 – May 2025

- Developed immersive VR and 3D interactive environments using Unreal Engine 5 and C++.
- Collaborated with the design team to integrate interactive mechanics, optimize performance, and refine user experience.
- Implemented gameplay logic, event triggers, and custom blueprint systems for advanced interactivity.

INFINIA Engineering | Unreal Engine Game Developer | Part-Time | Ankara — Sep 2024 – Jan 2025

- Focused on VR development, including object interaction systems and environment optimization.
- Designed interactive experiences compatible with multiple VR platforms.
- Worked closely with 3D artists and designers to implement visual storytelling within Unreal Engine.

INFINIA Engineering | Unreal Engine Game Developer Intern | Internship | Ankara — Jul 2024 – Aug 2024

- Assisted in developing gameplay systems, level blueprints, and interactive UI elements.
- Gained hands-on experience in game design, 3D modeling integration, and performance optimization.
- Learned best practices for collaboration in a production environment.

BTC Bilişim Hizmetleri | Intern Student | Internship | İstanbul — Jun 2024 - Jul 2024

- Worked on SQL-based data operations and SAP ERP systems.
- Contributed to database design and report generation.
- Assisted in improving internal workflows by leveraging ERP functionalities.
- Wrote and optimized SQL queries and built operational reports for internal business units.

SKILLS

- Game Development & Engines: Unity, Unreal Engine 5, PixiJS, HTML5 Canvas, VR, Multiplayer Networking.
- Languages: C#, C++, JavaScript/TypeScript, Python, SQL, Java.
- Practices & Tools: Object-Oriented Programming (OOP), Agile / Scrum, Jira, Git, Performance Optimization.
- Backend & Systems: Node.js, REST APIs, Docker, CI/CD, MongoDB.

LANGUAGES

English - Upper Intermediate
Turkish - Native

PROJECT

Project C-Building — Co-op Isometric Action Roguelite | Systems Programmer

- Tech: Unity, URP, C#, Netcode for GameObjects (NGO), UGS Relay/Lobby, OOP
- A 4-player co-op isometric action roguelite built in Unity, spanning 5 distinct biomes and a hero roster of 8 playable characters with fully composable ability kits.
- Architected a data-driven combat framework (ComposedAbilitySO pipeline) replacing hardcoded per-hero logic, enabling designers to compose abilities from reusable Effect/Delivery primitives while preserving hooks for bespoke hero mechanics.
- Built the multiplayer networking foundation using Unity Netcode for GameObjects (NGO) and UGS Relay/Lobby, including a session state machine, additive scene loading, and per-client camera isolation across 4 concurrent players.
- Designed a dual-camera system (fixed isometric + free-look "God's Eye" observer mode) driving the game's asymmetric finale sequence, where one player becomes a stationary tactical overseer for the remaining team.
- Implemented a dynamic, branching music system using DSP-time-anchored bar-aligned transitions across 5 biome soundtracks and a 3-state combat/escape sequence, ensuring seamless tempo-locked layer switching without desync.
- Collaborated directly with a game designer to translate evolving GDD/LDD specifications into versioned technical architecture, proactively flagging design-implementation gaps before development.

Bubbles — AI-Powered Turkish News Platform | Senior Capstone Project

Awards: Best Senior Project (System Development Award), CTIS Awards 2026
Best Presentation, Startup Studio Demo Day '26

- Tech: Node.js, Docker, MongoDB Atlas, Qdrant, Redis + BullMQ, Modal Labs (GPU), GitHub Actions
- Led the team's Agile / Scrum process — bi-weekly sprints, backlog grooming, and retrospectives — coordinating frontend, backend, and AI/ML developers in Jira.
- Architected and operated the backend: containerized microservices (Docker) on Railway, PM2 background workers for RSS ingestion and AI processing, and a Redis + BullMQ job queue.
- Built CI/CD pipelines with GitHub Actions automating build, test, and deployment; added health-check endpoints and external uptime monitoring after diagnosing a production outage.
- Managed MongoDB Atlas (indexing, TTL policies, performance tuning) and a Qdrant vector database powering multilingual semantic search.
- Integrated AI into production: a fine-tuned mBART Turkish summarizer and political / writing perspective-score models served on Modal Labs GPUs, with LLM taggers (DeepSeek, Groq) for enrichment.

BloomWake & AeroDrop — Browser-Based Games (CrazyGames)

- Tech: JavaScript, TypeScript, Canvas/WebGL, Vector Math
- BloomWake: Engineered a high-performance "Vampire Survivors" style bounded-swarm game with optimized rendering to handle hundreds of on-screen entities without frame drops.
- AeroDrop: Built a physics-based, cell-growing browser game featuring integrated water physics, a mass-based "Jet Boost" movement system, and dynamic bot AI simulation.

Not Enough Mana — 2D Browser-Based Card Game

- Tech: PixiJS, HTML5 Canvas, JavaScript, Game State Management
- Developed a 2D browser card game from scratch using PixiJS, implementing a modular graphics rendering pipeline, custom asset management, and robust turn-based game state logic.

Zombie Survival — Unreal Engine 5 (Co-op Prototype)

- Tech: UE5, Blueprints, Behavior Trees, NavMesh
- Prototyped a survival game loop; authored enemy AI using behavior trees and built scalable systems for pickups, combat, and inventory management.